IMDB Top 1000 Exploratory Data Analysis (SQL + Python)

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This project is aimed to explore trends and patterns within the IMDB top 1000 movies data set using SQL and Python based exploratory data analysis. My goal was to compare trends within movie genre, movie ratings and movie release year.

The analysis focuses on four main questions: -How do ratings compare among the highest-ranked movies? -Which years are most represented within top-rated films? -Which genre combinations achieve the highest average ratings? -How does genre popularity relate to average rating performance?

This notebook is a foundational piece for a follow up dashboard project that communicates the results in a more decision based format.

Dataset source: Kaggle IMDb Top 1000 Movies and TV Shows.

```
import pandasql as ps
```

```
df = pd.read_csv('/kaggle/input/imdb-dataset-of-top-1000-movies-and-tv-shows/imdb_top_1000.csv')
```

Top 20 rated Movies

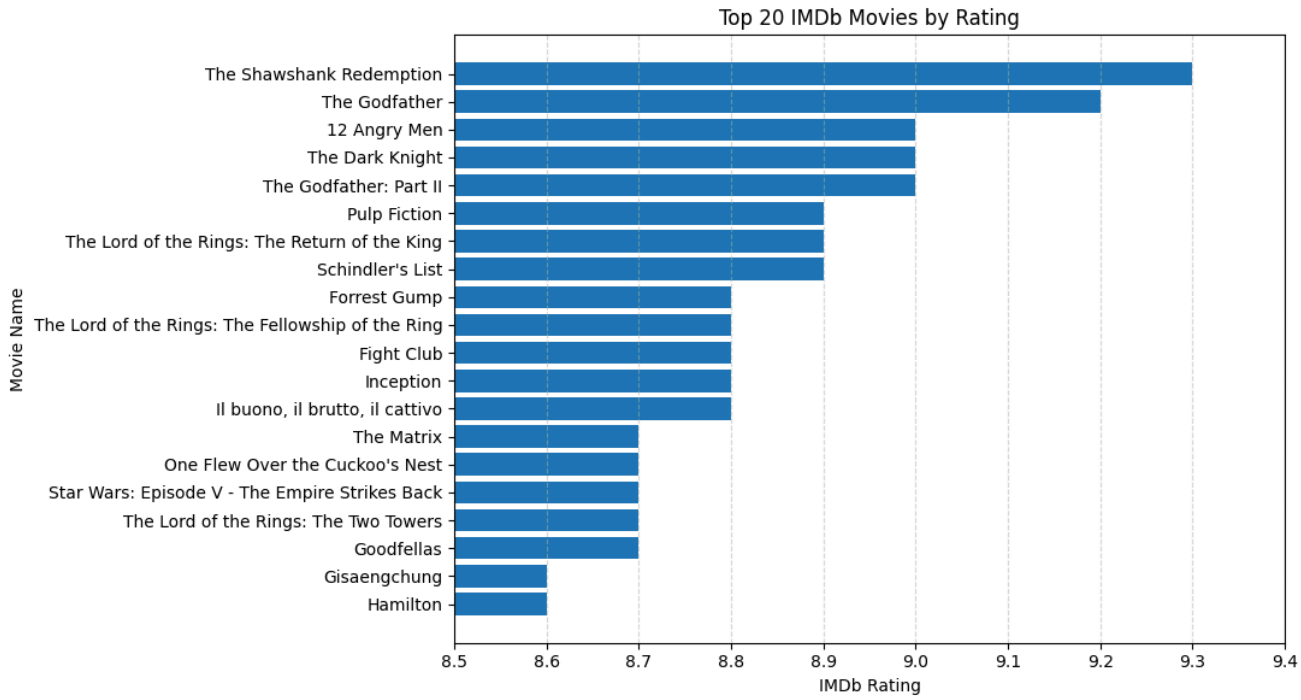
SQL Code

```
Top_20 = ps.sqldf("""
SELECT Series_Title AS "Movie Name", IMDB_Rating AS "Rating", Released_year AS "Year"
FROM df
WHERE released_year GLOB '[0-9][0-9][0-9][0-9]'
Order by Imdb_Rating DESC
Limit 20
""")
```

AI Assistance was used to support python visual code generation while maintaining full control over interpretation and insights.

Python Visual Code

```
import matplotlib.pyplot as plt
import numpy as np

Top_20['Rating'] = Top_20['Rating'].astype(float)
Top_20 = Top_20.sort_values('Rating')

plt.figure(figsize=(11,6))
plt.barh(Top_20['Movie Name'], Top_20['Rating'])

plt.title('Top 20 IMDb Movies by Rating')
plt.xlabel('IMDb Rating')
plt.ylabel('Movie Name')

plt.xlim(8.5, 9.4)
plt.xticks(np.arange(8.5, 9.4, 0.1))
plt.grid(axis='x', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
print("\n\n\n")
```



-Ratings among top 20 films are tightly clustered between 8.6-9.3, suggesting that even small rating disparities seperate elite titles.

-There is a noticable gap between the top 2 films (The Shawshank Redemption and The Godfather) and the rest indicating those movies consistently outperform in audience perception.

-Because the top movies cluster within a small scoring range, ranking order becomes more significant than actual ranking difference.

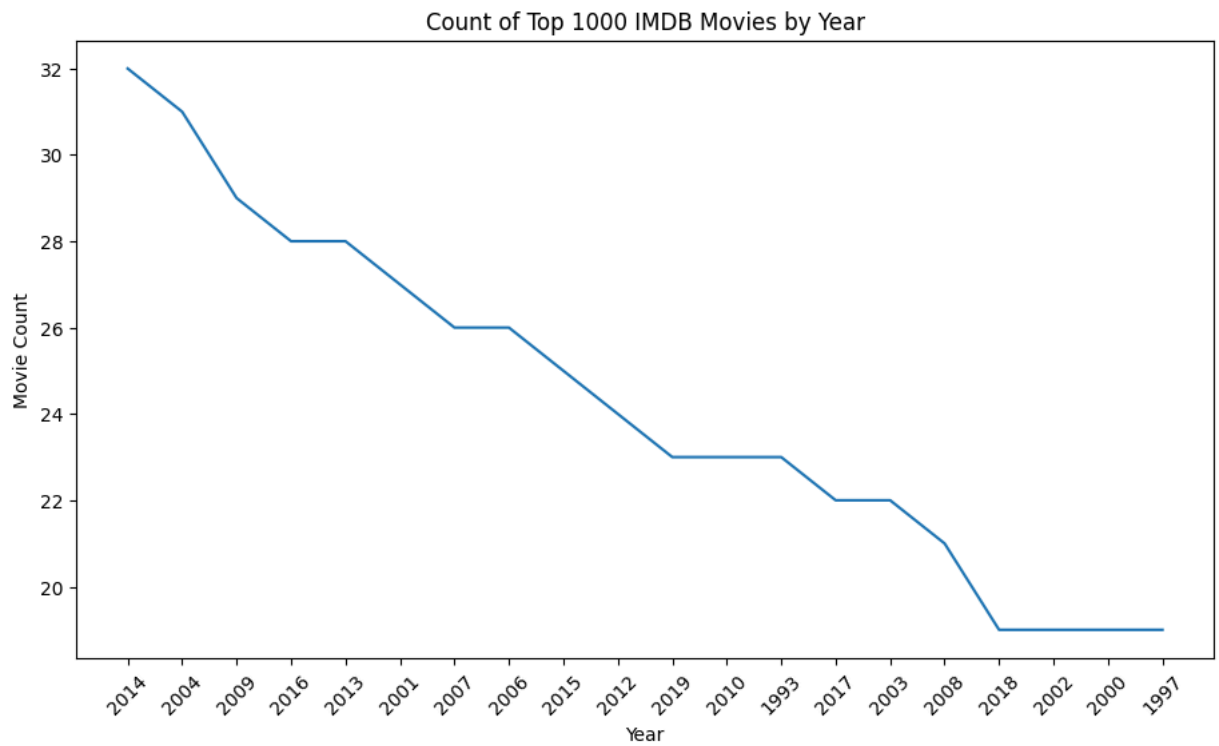
Years with the most top 1000 movies and their average ratings

SQL Code

```
Most_Top1000 = ps.sqldf("""
SELECT count(*) AS "Movie Count", released_year AS "Year", avg(imdb_rating) As "Average Rating"
FROM df
WHERE released_year GLOB '[0-9][0-9][0-9][0-9]'
Group by released_year
Order by "Movie count" desc, released_year desc
Limit 20
""")
```

Python visual code

```
import matplotlib.pyplot as plt

plt.figure(figsize=(11,6))
plt.plot(Most_Top1000['Year'], Most_Top1000['Movie Count'])

plt.title('Count of Top 1000 IMDB Movies by Year')
plt.xlabel('Year')
plt.ylabel('Movie Count')

plt.xticks(rotation=45)
plt.show()
print("\n\n\n")
```



-Movie count Indicates Strong presense of the early 2010s within the top 1000 list, which can suggest higher production volume, high engagement levels or recency bias.

-Since only top 1000 movies are recorded, this trend represents elite movies quality, not overall yearly movie quality.

-Older films could be underrepresented due to historical differences in audience size and rating accessibility.

Top 30 Highest Rated Genre Combinations

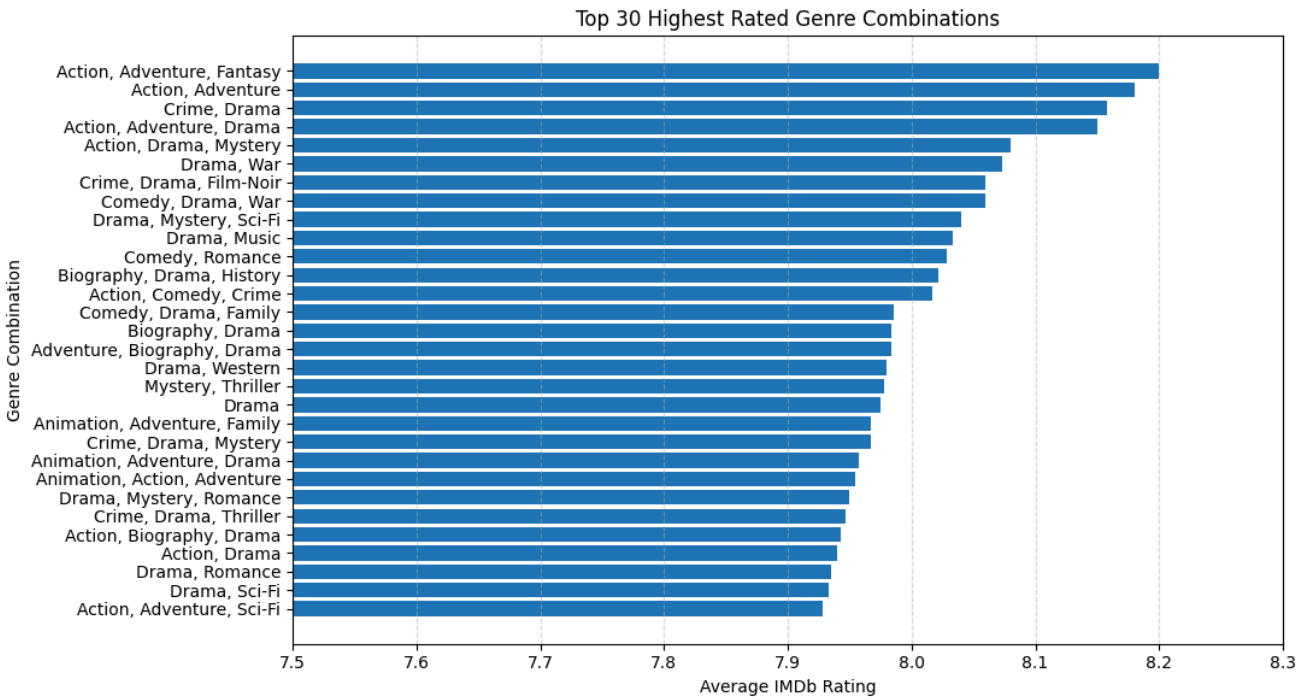
SQL Code

```
HRGC = ps.sqldf("""
SELECT Genre, Count(*) AS "Movie Count", avg(IMDB_Rating) AS "Average Rating"
From df
Group by Genre
Having count(*) >=5
Order by "Average Rating" Desc, "Movie count" DESC
Limit 30
""")
```

Python Visual Code

```
import matplotlib.pyplot as plt

# make sure rating is numeric
HRGC["Average Rating"] = HRGC["Average Rating"].astype(float)

# sort so highest rating appears at top
HRGC = HRGC.sort_values("Average Rating")

plt.figure(figsize=(11,6))

plt.barh(HRGC["Genre"], HRGC["Average Rating"])
```

```
plt.title("Top 30 Highest Rated Genre Combinations")
plt.xlabel("Average IMDb Rating")
plt.ylabel("Genre Combination")

# cleaner x-axis
plt.xlim(7.5, 8.3)
plt.grid(axis='x', linestyle='--', alpha=0.5)

plt.tight_layout()
plt.show()
print("\n\n\n")
```



-Multiple genre movies consistently outperform single genre movies indicating films that blend elements are more successful.

-Drama is the lone single genre appearing in the top combinations indicating it is a strong foundation for movie production.

-Genre diversity may increase appeal by widening their influence across multiple audience preferences.

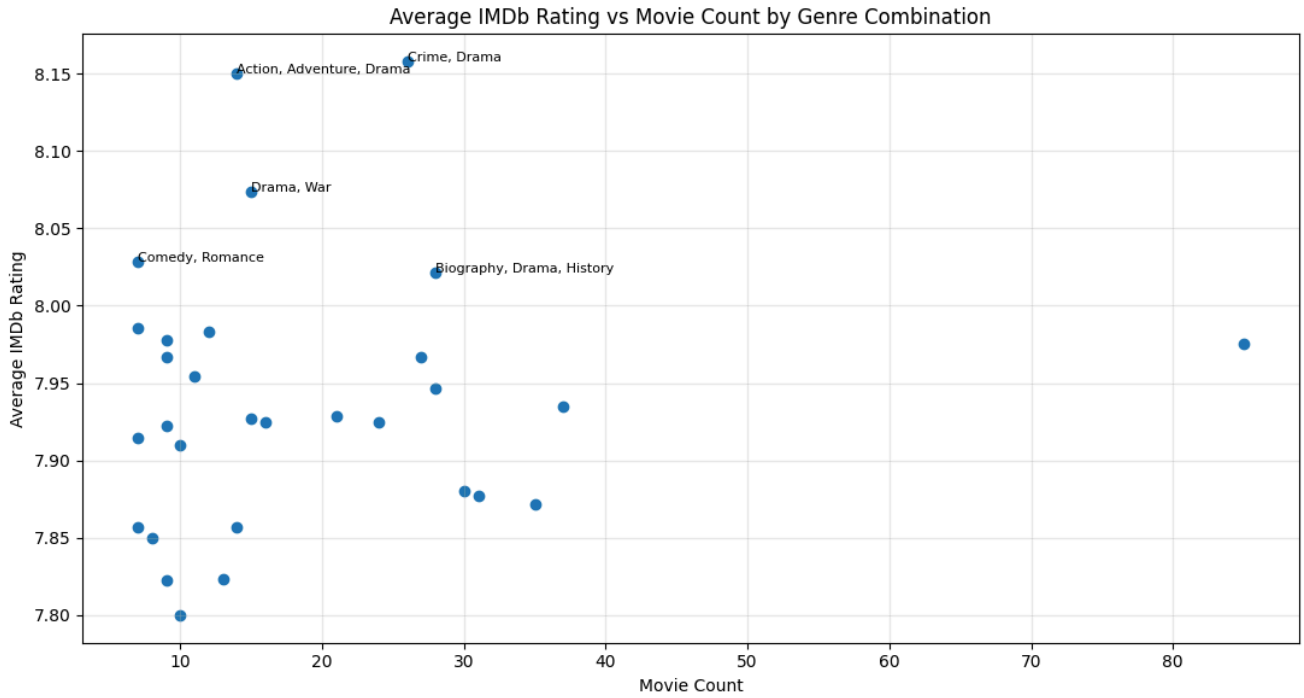
Average Rating vs Movie Count

SQL Code

```
ARvMC = ps.sqldf("""
SELECT count(Series_Title) AS "Movie Count", Avg(IMDB_Rating) AS "Average Rating", genre
from df
Group by genre
Having count(*) >= 5
Order by "Movie Count" desc
Limit 30

""")
```

Python Visual Code

```
import matplotlib.pyplot as plt

plt.figure(figsize=(11,6))

plt.scatter(
    ARvMC["Movie Count"],
    ARvMC["Average Rating"]
)

# label only higher-rated genres
for i, genre in enumerate(ARvMC["Genre"]):
    if ARvMC["Average Rating"].iloc[i] >= 8.0:
        plt.annotate(
            genre,
            (ARvMC["Movie Count"].iloc[i],
             ARvMC["Average Rating"].iloc[i]),
            fontsize=8
        )

plt.title("Average IMDb Rating vs Movie Count by Genre Combination")
plt.xlabel("Movie Count")
plt.ylabel("Average IMDb Rating")

plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
print("\n\n\n")
```



-High-volume genres cluster around slightly lower averages, indicating that mainstream categories produce more variability in quality.

-The outlier with high movie count but strong rating suggests certain genres maintain both popularity and consistency.

Final Insights

-Ratings among the top 20 films are tightly clustered between 8.6 and 9.3, showing that small rating differences separate elite titles and make ranking order more meaningful than large differences in quality perception.

-The early 2010s show strong representation within the Top 1000 list, which may reflect higher production volume, increased audience engagement, or recency bias in online ratings.

-Because the dataset only contains top-ranked films, trends observed represent characteristics of elite movies rather than overall yearly film quality, and older films may be underrepresented due to historical differences in audience size and rating accessibility.

-Genre combinations consistently outperform single-genre films, suggesting that blending multiple genres increases audience appeal and overall rating performance.

-Drama appears as the only single genre within the highest-rated combinations, indicating it may serve as a strong foundational genre when paired with others.

-High-volume genres tend to cluster around slightly lower average ratings, while niche genres can achieve higher averages, suggesting a tradeoff between popularity and rating consistency.